Out-of-Core, Tile-Parallel Hyperspectral Processing on Commodity Hardware: an OpenMP/OpenCL Engine Integrated with QGIS

Aaron R
Digital University Kerala
Thiruvananthapuram, Kerala, India
aaronr43210@gmail.com

Abstract—Spaceborne imaging spectrometers such as EnMAP deliver scenes of hundreds of spectral bands; a single 224-band EnMAP L2A scene used here is 686 MB and holds about 2.4×10^8 reflectance values. Desktop GIS tools typically process such cubes in memory, which limits them on laptops. We present hyperspatial_cli, a C++17 engine that decomposes a scene into independent tiles and processes them out of core: OpenMP threads with dynamic scheduling and thread-private GDAL handles compute seven spectral indices, a two-pass principal component analysis and the Spectral Angle Mapper, and a portable OpenCL 1.2 backend runs the same kernels on GPUs of any vendor with a per-tile CPU fallback. The engine is driven from QGIS by a binary-free plugin that never blocks the user interface, and its GeoTIFF tiles are the unit of distribution for downstream Apache Spark analytics. On a 12-core Apple M4 Pro all seven indices of the whole scene take 2.02 s (256-px tiles, 8 threads), SAM 0.99 s and PCA 5.96 s, with peak memory set by threads and tile size rather than scene size. A tile-size sweep shows that granularity limits scaling: the 8-thread speed-up rises from 1.2× with 4 tiles to 5.7× with 100 tiles, until per-tile overhead dominates. CPU and GPU outputs agree to float rounding, except one pixel in 1.5×10^6 for two indices. The GPU backend is not faster on this machine: the workload is I/O-bound and the driver forces a single host thread in GPU mode. We report these measurements and the design lessons they imply.

Index Terms—hyperspectral imaging, OpenMP, OpenCL, out-of-core processing, heterogeneous computing, GIS, EnMAP

I. Introduction

Imaging spectroscopy missions (EnMAP [1], PRISMA, EMIT) produce cubes whose size grows with the number of bands rather than only with area. Most users of these data work in desktop GIS, where open-source hyperspectral tools commonly hold the cube in memory, and GPU acceleration is mainly found in commercial or NVIDIA-specific software [2]. Portable OpenCL implementations of hyperspectral kernels have been studied as research prototypes [3]; our focus is an engineering one: making full-scene processing fast, memory-bounded and robust on the hardware GIS users already own, inside the GIS. Contributions: (i) an out-of-core, tile-parallel engine with a closed-form memory model; (ii) a vendor-portable OpenCL backend with per-tile CPU fallback; (iii) a measured evaluation on a real EnMAP scene, including a negative GPU result and its cause.

II. Engine Design

A. Decomposition and out-of-core streaming

A tiler splits the $W \times H \times B$ scene ($1259 \times 1216 \times 224$) into square tiles of side s (default 512 px, 9 tiles). For the index pipeline tiles are independent, so the problem is embarrassingly parallel. Each worker reads one tile, runs a six-stage pipeline (read, no-data handling, bad-band masking, indices, per-band normalisation, write) and writes one GeoTIFF with $B+7$ bands. The scene is never resident; with T threads the working set is

$$M \approx T\, s^2 B \cdot 4 \text{ bytes } (\approx T \times 235\,\text{MB at } s = 512), \quad (1)$$

independent of scene size.

B. Shared-memory parallelism

The tile loop is an OpenMP parallel for with schedule(dynamic), because edge tiles of a rotated satellite footprint carry far less work (up to 97 % background). GDAL datasets are not thread-safe, so every read opens a thread-private handle and every tile is written to its own file; the only shared state is the first error, guarded by an atomic flag and a mutex. No exception may leave the parallel region: each iteration catches, records and continues, and the process exits with code 2.

C. Heterogeneous backend

The OpenCL 1.2 backend selects a GPU device on Apple, NVIDIA, Intel or AMD platforms. Its index kernel is a line-for-line port of the CPU code; SAM [4] and PCA have dedicated kernels, the latter with Kahan-compensated float32 reductions. Any initialisation or dispatch failure falls back to the CPU for that tile. Because the Apple OpenCL queue was observed to corrupt results when fed from several host threads, GPU mode runs the tile loop on a single host thread.

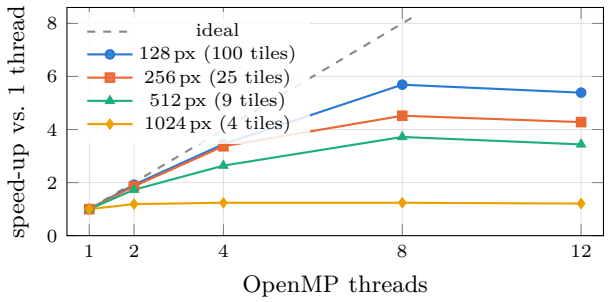


Fig. 1. Strong scaling of the seven-index pipeline on the whole scene for several tile sizes. Smaller tiles expose more parallelism.

TABLE I
Whole-scene runtime, memory and CPU–GPU agreement

Operation	1 thr.	best CPU	GPU	max |CPU − GPU|
Indices (7)	8.60 s	2.52 s	9.13 s	$\leq 1.5\times10^{-5}$ [a]
SAM	—	0.99 s	1.15 s	1.1×10^{-6} rad
PCA (3 PCs)	—	5.96 s	5.79 s	1.6×10^{-5} of range
Peak memory	1.38 GB	4.21 GB	1.39 GB	

[a] Except one pixel in 1.5×10^6 for NDVI and NDWI (zero-operand edge case). 512-px tiles; best CPU: 8 threads (indices), 12 threads (SAM, PCA); from engine_benchmark.csv. Fig. 1 uses a separate sweep (engine_tile_sweep.csv); single-thread times differ by up to 9 % between the two sessions.

D. Numerical robustness and integration

PCA uses two passes in double precision (exact means, then $\Sigma(x-\mu)(x-\mu)^T$), a Jacobi eigensolver with a deterministic sign convention, and a projection pass. Bands are selected by wavelength from file metadata. A QGIS plugin launches the engine as a subprocess from a worker thread, streams its progress and loads styled layers; the tiles feed Apache Sedona jobs on Spark unchanged, complementing cluster-scale Spark processing of hyperspectral data [5].

III. Evaluation

Setup. EnMAP L2A scene (Tyrolean Alps, 12 June 2022); Apple M4 Pro (8 performance + 4 efficiency cores, 24 GB); engine time from its RESULT line, median of three runs.

Scaling. Fig. 1 shows that tile granularity, not core count, limits scaling. With the default 512-px tiles the scene has only 9 tiles, which cannot be balanced over 8 threads (one thread processes two), so 8 threads give 3.7× (46 % efficiency). Finer tiles expose more parallelism: 4.5× with 256-px tiles (25 tiles) and 5.7× with 128-px tiles (100 tiles), whereas 1024-px tiles (4 tiles) saturate at 1.2×. Finer tiles also add per-tile overhead (one thread needs 20.5 s at 128 px against 9.1 s at 256 px), so the fastest configuration is 256-px tiles on 8 threads: 2.02 s for all seven indices, 20 % faster than the 512-px default and with 2.73 instead of 4.15 GB peak memory. No configuration improves beyond 8 threads; we attribute this to the four efficiency cores and to I/O (reading 686 MB, LZW-compressing 1 GB), to be confirmed by an instrumented read/compute/write breakdown.

Memory. Peak resident memory grows with threads (1.38 GB serial, 2.46 GB with 2, 4.21 GB with 8 threads) and stays bounded because at most one tile per thread is in flight, matching Eq. (1).

CPU vs. GPU. Table I: the GPU path reproduces the CPU results but is not faster. Indices perform a few operations per value, so time is dominated by reading the 686 MB cube and compressing 1 GB of output; in GPU mode that I/O runs on one host thread, and the GPU run (9.13 s) matches the serial CPU run (8.53 s). For SAM and PCA the GPU is on par with 12 CPU threads, while SAM's GPU run needs 0.67 GB instead of 2.53 GB.

IV. Conclusion and Future Work

A tile-parallel, out-of-core design processes a complete 224-band scene in seconds on a laptop with bounded memory, and a portable OpenCL backend reproduces the CPU results on a non-NVIDIA GPU. The measurements point to the next steps: a 256-px default tile size, a pipelined hybrid mode (multithreaded tile I/O feeding one GPU dispatch thread), evaluation on x86 servers with discrete GPUs, weak scaling over multi-scene mosaics, and distribution of tiles across a Spark cluster.

References

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